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Frog in the well: A review of the scientific literature for evidence of amphibian sentience

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ABSTRACT

Millions of amphibians are traded annually around the world for the exotic pet industry. Their experience during both trade, and in captivity as pets, leads to numerous animal welfare issues. The poor welfare of many pet amphibians is due in part to poor attitudes and acceptance of amphibian species to suffer. Amphibians, like other vertebrate species, are sentient, which means that their feelings matter. In our study, we have sought to explore the scientific literature over 31 years (1990–2020), to establish what aspects of sentience are accepted and still being explored in amphibians. Our review aimed to; 1) assess the extent to which amphibian sentience features in a portion of the scientific literature, 2) to determine which aspects of sentience have been studied in amphibians, and in which species, and 3) to evaluate what this means in terms of their involvement and treatment in the global exotic pet trade. We used 42 keywords to define sentience and used these to search through four databases (ScienceDirect, BioOne, Ingenta Connect, and MDPI), and one open-access journal (PLOS ONE). We recorded studies that either explored or assumed sentience traits in amphibians. We found that amphibians were assumed to be capable of the following emotions and states; stress, pain, distress, suffering, fear, anxiety, excitement, altruism and arousal. The term 'emotion' was explored in amphibians with mixed results. Our results show that amphibians are known to feel and experience a range of sentience characteristics and traits and that these feelings are utilised and accepted in studies using amphibians as research models. There is, however, still much more to learn about amphibian sentience, particularly in regards to positive states and emotions, and this growing understanding could be used to make positive changes for the experiences of amphibians in captivity.

1. Introduction

Sentience is the capacity to feel and experience a range of subjective states and experiences, whether they are positive or negative (Birch, 2017; Proctor, 2012). Although vertebrates are generally accepted to be sentient, some taxonomic groups such as insects (Lambert et al., 2021), reptiles (Lambert et al., 2019), and amphibians (Ferrell, 2021) are used or traded extensively, often with insufficient knowledge of their welfare needs.

Amphibians are a diverse, colourful and fascinating taxonomic group. Their ability to metamorphose, inhabit both terrestrial and aquatic environments, and their unique range of phenotypes, have led them to be an increasingly popular exotic pet around the world (Altherr and Lameter, 2020; Burghardt, 2017; Measey et al., 2019). In fact, the international exotic pet market is considered to be a key driver of the global wildlife trade in amphibians with millions traded around the world as pets every year (Harrington et al., 2021). Amphibians are also heavily exploited for use in biomedical research, traditional medicines, and consumption (Gerson, 2012; Hughes et al., 2021; Little et al., 2021).

The commercial trade in amphibians is comprised of both legal and illegal markets (Macdonald et al., 2021). Commercial trade of wildlife is reported to deliver societal and economic advantages by providing livelihood opportunities (Harrington et al., 2021) and potential conservation benefits such as improved protected areas and enhanced efforts to tackle illegal trade (Cooney et al., 2018; Roe and Lee, 2021), particularly in biodiverse developing countries. Yet, trade is also

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associated with several negative impacts, including the potential spread of invasive species and infectious diseases (Can et al., 2019; O'Hanlon et al., 2018). The global amphibian trade can also put substantial pressure on wild populations, particularly when they are already under threat (Harrington et al., 2021; Hughes et al., 2021). From the individual animal's perspective, the treatment of amphibians during the trade process (e.g., capture, restraint, captive breeding, and transportation), and their subsequent experience of captivity, also poses numerous animal welfare issues of concern (Baker et al., 2013). Being seized, transported and kept in a captive environment is highly likely to cause any sentient animal some degree of stress, particularly as amphibians have specialist needs in terms of air quality, temperature, humidity and diet (Grant et al., 2017).

1.1. Study aims

Despite their popularity as exotic pets, amphibians are often viewed as less capable of emotions compared with other taxa, such as mammals (Callahan et al., 2021; Higgs et al., 2020). A lack of appreciation of the complexity of amphibians and their capacity to feel and suffer has important implications for how they are treated (Burghardt, 2013). To be able to mitigate any negative animal welfare implications associated with the amphibian trade, an understanding of the full landscape of amphibian sentience is required, as is the need to identify areas of strength and clarity in terms of what we do know about amphibian sentience as well as areas where more research is required. In this study, we have sought to review the scientific evidence and understanding of amphibian sentience within the last 31 years (1990–2020). The specific aims of our review were to; 1) assess the extent to which amphibian sentience features in a portion of the scientific literature, 2) to determine which aspects of sentience have been studied in amphibians, and in which species, and 3) to evaluate what this means in terms of their involvement and treatment in the global exotic pet trade.

2. Methods

2.1. Keywords

To search for evidence and the utilisation of sentience traits in amphibians in the scientific literature, we used 42 keywords that referred to traits and aspects of animal sentience (Appendix 1). The keywords were taken from a list used in a previous review into animal sentience (Proctor et al., 2013). The original list of 174 keywords used in 2013, has since been used in two subsequent reviews; one focused on reptiles (Lambert et al., 2019), and another focused on insects (Lambert et al., 2021). In this current review, we focussed our efforts on the keywords that had returned results in the previous three reviews. As with the past two reviews, we removed the keywords that referred to cognitive abilities and personality, as these were not in line with the aims of the current review. We also added two additional keywords that we had not used before, 'intentionality' and 'prosocial', as we felt that they would benefit the review, and help us to encapsulate additional aspects of sentience.

2.2. Literature search

The 42 keywords were used to search through four databases (ScienceDirect, BioOne, Ingenta Connect, and MDPI), and one open-access journal (PLOS ONE). The searches were performed with each of the amphibian orders, using the order's common name (frog, salamander and caecilian). To collate as many articles as possible, we broke down the order 'frog' further into frog and toad, and the order 'salamander' into salamander and newt, and then removed any duplicates. The Boolean operator AND was used to connect the keyword and the amphibian order, for example, we searched for pain AND frog. We searched for articles published within the timeframe of 1990–2020, as this allowed us to look at the trends in publications over the past 31 years, and to look at a large and recent dataset.

In total, we performed 1050 searches using each of the keywords along with the different orders, across the databases. To be included in the review, each of the returned articles had to (1) be a research article, and not a review paper, (2) be utilising an amphibian species in their experiment, (3) be using the keyword in relation to the amphibian being studied, and not in reference to another study's findings, and (4) use the keyword to refer to the amphibian's emotional state (e.g. the mental state of distress rather than distress calls). If all four criteria were met, then the article was included in the review. Each result was then categorised as either 'Assume' or 'Explore', depending on how the keyword was used. If for example, the authors of the study were assuming the frog in question could feel pain and were using this understanding to test analgesics, then the result was marked as Assume. However, if the study explored whether or not a species of newt was capable of demonstrating empathy, then this would have been marked as 'Explore'. The 'Explore' results were then categorised as either 'Positive' or 'Negative' depending on whether the article successfully found evidence of the keyword, or not. Other details about the article were also recorded, including the article title, the journal it was published in, and the publication year.

If the article referred to more than one amphibian species, each species was treated separately, and only included if the keyword was used in reference to them. In some articles, different keywords were used with different species, and so were recorded as such. In others, multiple amphibian species were utilised, but only one or two were referred to in relation to the keyword. In those cases, only the species with which the keyword was used in conjunction were included. The keywords 'stress' and 'suffering' returned many articles that referred to the adherence to ethical procedures. Only articles that specifically referred to 'suffering' or 'stress' in the amphibian species were included, and those that referred to the terms only in regards to the ethics procedures were excluded. For example, the phrase "care was taken to minimise suffering in the research animals" was used repeatedly. These articles did not refer specifically to suffering in the amphibian species and may have referred to other species used in the study, and so were therefore excluded. Whereas articles that mentioned the species directly were included. For example, when a study stated that they "performed euthanasia on the frogs to cease their suffering", the authors were assuming that the frogs were capable of suffering, and so the article met the criteria for inclusion. Following the initial searches, one researcher reviewed each of the returned articles against the four criteria to determine which articles to include in the review.

2.3. Data analysis

Descriptive analyses were performed on the returned results using Microsoft Excel Version 2109.

3. Results

A total of 189 results were returned, across 150 unique articles. The results came from 56 different journals across the databases searched. Of the 150 articles, 88 (58.7%) of them came from the database Science Direct, 25 (16.7%) came from BioOne, 16 (10.7%) from Ingenta Connect, 21 (14%) from the journal PLOS ONE, and none came from the journal group MDPI. The three journals with the most returned articles were PLOS ONE (n = 21, 14% of all articles), Hormones and Behaviour (n = 13, 8.7%), and General and Comparative Endocrinology (n = 11, 7.3%).

Fig. 1 shows the number of returned articles by publication year. Articles were returned for each of the publication years reviewed (1990–2020). Overall, our results show an increasing trend ($R^2 = 0.65$) in the number of articles published each year since 1990 that referred to the sentience keywords in relation to amphibians. For example, there were 16 articles published in 2020 which referred to a sentience trait in amphibians, compared with just two in 1990.

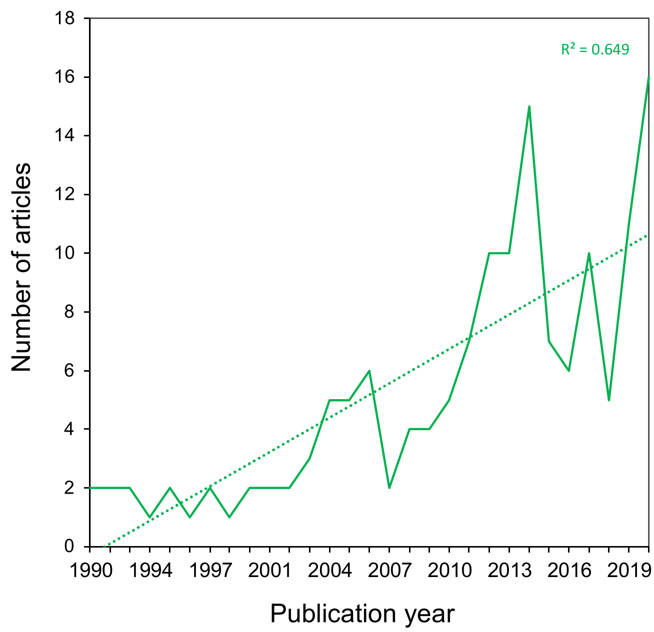

Fig. 1. The number of articles published each year using the sentience keywords.

3.1. *Sentience keywords returned*

Of the 42 keywords we searched for, 10 (23.8%) returned results: 'Stress', 'Pain', 'Distress', 'Suffering', 'Emotion', 'Fear', 'Anxiety', 'Excitement', 'Altruism' and 'Arousal' (see Fig. 2). The top three keywords returned were 'Stress' (n = 119, 62.9% of all results), 'Pain' (n = 27, 14.2%), and 'Distress' (n = 13, 6.8%).

3.2. Taxa returned

Of the three amphibian orders we searched for, all returned results (Anura (e.g. frogs) n = 125 results, 66.1% of all results; Gymnophiona (i.e. limbless caecilians) n = 4, 2.1%; Urodela (i.e. newts, salamanders) n = 60, 31.7%). Sentience related keywords had been assumed for Anura in 117 articles, explored with a positive outcome in two articles and with a negative outcome in four articles (Fig. 3). Sentience related keywords had been assumed in 63 and four articles for the orders Urodela and Gymnophiona, respectively (Fig. 3).

Of the 10 keywords returned, all were assumed apart from the keyword 'Emotion', which was explored (Fig. 4). 'Emotion' was explored in six results (3.1% of all results), comprising six different frog species and sub-species (6.8% of all species). The species were; American bullfrogs (*Rana catesbeiana*), cane toads (*Rhinella marina*), European fire-bellied toads (*Bombina bombina*), leopard frogs (*Lithobates pipiens*), Northern leopard frogs (*Rana pipiens*), and poison dart frogs (*Dendrobates tinctorius*). Two of the explored results had a positive outcome (for poison dart frogs and leopard frogs), and four had a negative

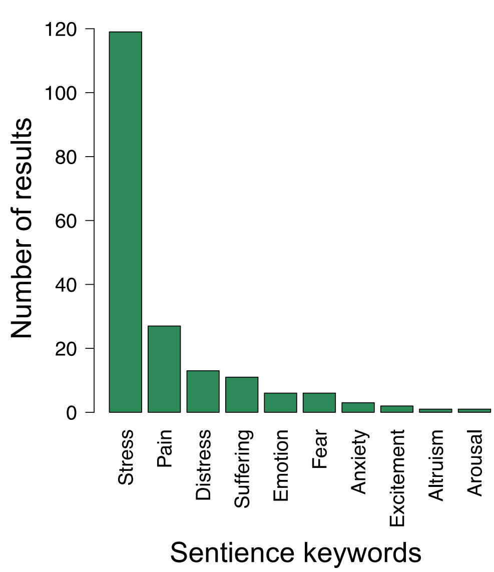

Fig. 2. The number of returned articles across the sentience keywords searched.

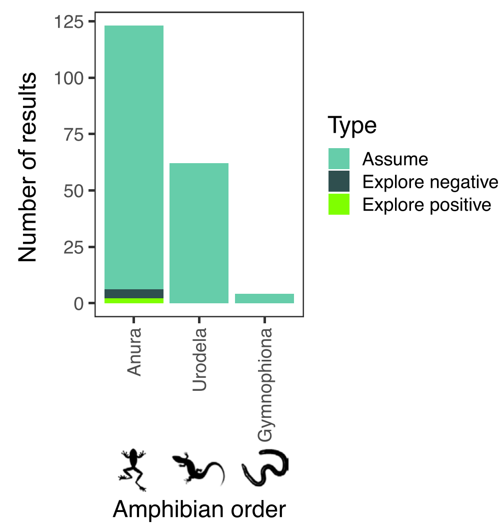

Fig. 3. The number of returned articles per keyword for the three different amphibian orders. Shown is the number of articles returned that either assumed amphibian sentience, explored amphibian sentience with a positive outcome, or explored amphibian sentience with a negative outcome.

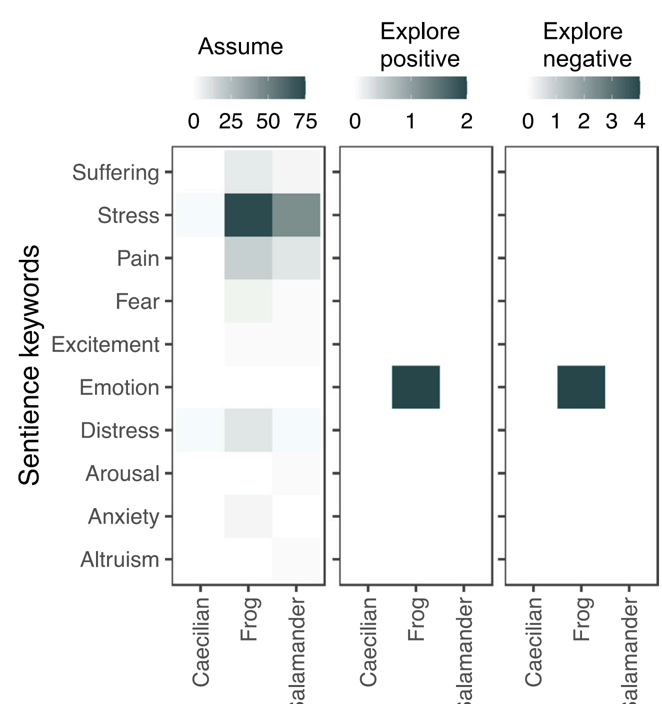

Fig. 4. Summary of the sentience keywords that returned results. The frequency of the keyword returns for each type of amphibian are shown, along with whether the keyword was a) assumed, b) explored with a positive result, or c) explored with a negative result.

outcome (for American bullfrogs, cane toads, European fire-bellied toads, and Northern leopard frogs).

Twenty-three different amphibian taxonomic familiae were featured in the 150 articles returned. Of these, 17 (73.9%) were frog families, including three (13%) toad families, four (17.3%) were salamander families, of which one (4.3%) was specifically newts, and two (8.6%) were caecilian families (Fig. 5).

The returned articles featured a total of 87 different species and sub-species of amphibians (see Appendix 2). Of these, 40 (45.9% of all species) were frogs, 16 (18.3%) were toads, 20 (22.9%) were salamanders, 9 (10.3%) were newts, and two (2.2%) were caecilians. The most frequently featured species overall were the Northern leopard frog (*Rana pipiens*) (n = 14, 7.4% of total results), cane toad (*Rhinella marina*) (n = 13, 6.9%), African clawed frog (*Xenopus laevis*) (n = 10, 5.3%), and rough-skinned newt (*Taricha granulosa*) (n = 8, 4.2%). Fifteen (7.9%) of the articles returned more than one amphibian species, whereas the remaining 135 (71.4%) only referred to one species.

4. Discussion

Like all vertebrates, amphibians are considered scientifically to be sentient, feeling beings (Jones, 2012). This means that for them, their feelings matter, and this has a direct consequence on how we treat and use them, as well as for their performance and survivability in commercial trade and in research (Baker et al., 2013). In our review of the scientific literature, 10 of the 42 sentience keywords we searched for returned results (stress, pain, distress, suffering, emotion, fear, anxiety, excitement, altruism and arousal), featuring 23 different amphibian taxonomic families (Fig. 5) and 87 different species and sub-species (Appendix 2). Therefore, these particular sentience traits appear to be widely accepted across a diversity of amphibian taxa. Furthermore, our findings show an increasing trend (R = 0.65) in the number of scientific publications assuming and exploring sentience in amphibians over the past three decades (Fig. 1).

4.1. Anura (frogs)

Of the three amphibian orders we searched for, Anura (frogs) was the most frequently returned. Anura featured in 66.1% (n = 125) of the returned results, involving 40 different species and sub-species. Eight sentience keywords were returned for Anura: anxiety, distress, emotion, excitement, fear, pain, stress, and suffering. All of these keywords, apart from emotion, were ‘assumed’ in the 125 returned results. These experiences and feelings are, therefore, widely accepted in frogs, and are used and accounted for in experiments involving them. For example, the term ‘stress’ was often used in reference to a frog’s experience during an experimental procedure (e.g. McDaniel et al., 2008).

‘Emotion’ was the only keyword that produced mixed results in regards to identifying sentience in amphibians, as two returned articles reported a negative result in four different species (Cabanac and Cabanac, 2000, 2004). In these studies, Cabanac suggested that the lack of a stress tachycardia or emotional fever response to the unpleasant experience of handling indicated that the frog species studied (cane toad, *Bufo marinus,* European fire-bellied toad, *Bombina bombina,* American bullfrog*, Rana catesbeiana and* Northern leopard frog*, Rana pipiens)* were incapable of emotional experience. However, the link between these phenomena and both consciousness and emotional states is not conclusive (Burghardt, 2013; Jacobson et al., 2021). Furthermore, a recent study has shown that manually restraining poison dart frogs and leopard frogs elicits stress tachycardia in both species (Jacobson et al., 2021), suggesting that the initial gentle handling stimulus used by Cabanac (2000, 2004) was not stressful enough to elicit a response, and that their findings were not necessarily indicative of a lack of emotion in amphibians (Jacobson et al., 2021).

4.2. Urodela (salamanders)

The order Urodela (salamanders) returned seven keywords: altruism, arousal, distress, excitement, pain, stress and suffering. These keywords were ‘assumed’ in 61 different species and sub-species of Urodela, and returned 62 results overall. Altruism, for example, was assumed in smooth newts (*Lissotriton vulgaris*) and was used to explore the role of kin recognition in egg-cannibalism (Tóth et al., 2011). As with the order Anura, the keyword ‘stress’ was the most frequently used keyword for Urodela (45 results and species). In one study, male spotted salamanders (*Ambystoma maculatum*) were subjected to handling and restraint to determine what effect their capacity to feel mental stress would have on their sexual behaviour and reproductive abilities (Woodley and Porter, 2016).

4.3. Gymnophiona (caecilians)

For the order Gymnophiona, two species of caecilians were featured in our results: the Mexican burrowing caecilian (*Dermophis mexicanus*) and the Koh Tao Island Caecilian (*Ichthyophis kohtaoensis*). Both species of caecilians featured in one study, where they were abruptly exposed to microgravity to determine their behavioural reactions (Wassersug et al., 2005). The caecilian species were assumed to be capable of distress and stress, and signs of these subjective states were looked for during the parabolic flight and were accounted for during the preparation for the experiment. Although Gymnophiona only featured in one of the studies covered in our review, this may reflect their relative absence in captivity, compared with other amphibian orders, rather than a reflection of their capacity for sentience. Fig. 6.

4.4. What do the results mean for the trade in amphibians?

Millions of amphibians are traded internationally every year, and analyses suggest that over 1200 species are currently involved, representing 17% of all known amphibian species on Earth (Hughes et al., 2021). The number of amphibian species affected by trade is also predicted to increase. For example, Scheffers et al. (2019) estimated that more than 4900 amphibian species may be traded in the future. Amphibians traded legally under the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) only represent a small fraction of the species being traded globally, and the true scale of the amphibian trade is largely unquantified (Bush et al., 2014; Hughes, 2021; Lockwood et al., 2019). What is known is that the numbers are huge. Of the CITES regulated amphibian species, the US is the largest

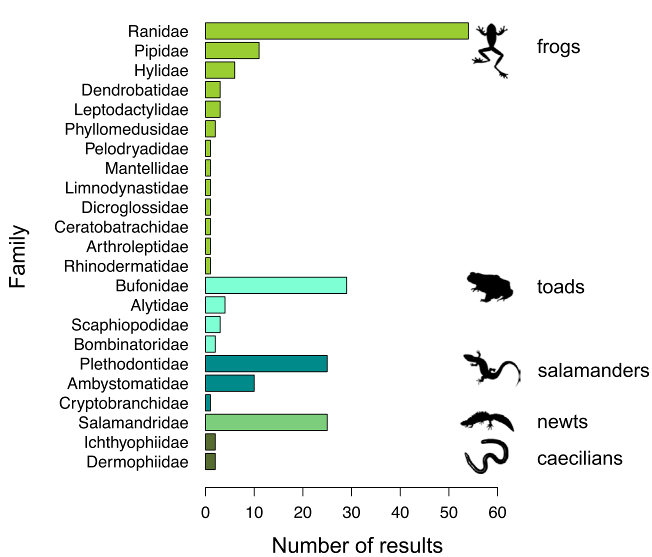

Fig. 5. The number of returned articles by the amphibian family.

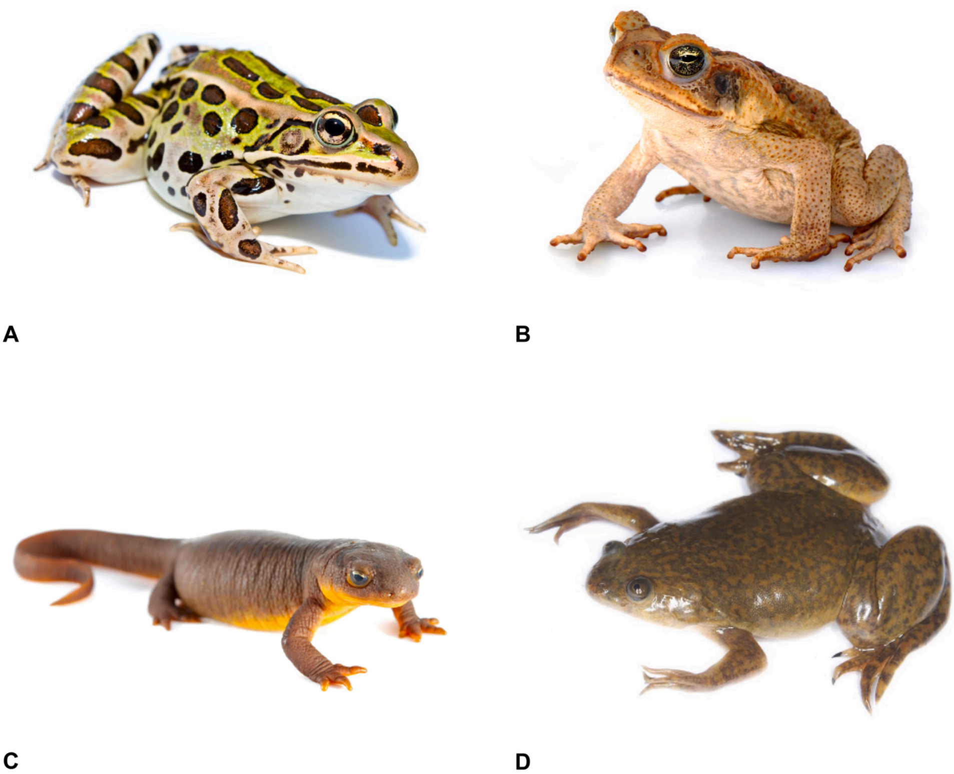

Fig. 6. Some of the most frequently returned amphibian species. A, Northern leopard frog (*Lithobates pipiens)* by Brian Gratwicke, licensed under CC BY 2.0; B, African clawed frog (*Xenopus laevis*) by USFWS Fish and Aquatic Conservation, licensed under CC BY 2.0; C, Rough-skinned newt (*Taricha granulosa*) by Nature Picture Library / Alamy Stock Photo; and D, Cane toad (*Rhinella marina*) by Sam Fraser-Smith, licensed under CC BY 2.0.

importer; between 2012 and 2016 the US imported more than 100,000 individual amphibians from 55 different species and 17 different countries (Can et al., 2019). In the same period, Nicaragua was the largest exporter of CITES-listed amphibians, exporting over 122,000 amphibians from 2 species, to 20 different countries worldwide (Can et al., 2019). Newly identified species are also very quickly sought-after in the international exotic pet trade before anything is known about their wild population status, or their suitability to captivity (Altherr and Lameter, 2020).

Amphibians are unique in their behaviour and needs, and as a result, they can suffer greatly throughout the trade chain (Grant et al., 2017). For example, frogs are reportedly collected from the wild by hand at night, located by their calls, and are often placed into cloth sacks or bags along with others of varying species. Captured individuals may then be left in buckets for long periods of time, in numbers of up to a hundred, with only a little water to keep them moist (D'Cruze, personal observation). Capturing amphibians in this way pays little if any regard to the sentience of these animals, is likely to cause extensive suffering, and risks disease transmission between individuals. During the transportation process, it has been reported that large volumes of animals can be crammed into small containers and taken on long journeys, placing them at risk of overcrowding, disease, crushing, injuries, and painful, prolonged deaths (see Ashley et al., 2014; Harrington et al., 2021 for examples).

An investigation of a major exotic companion animal wholesaler found that around 44.5% of over 26,400 individual amphibians confiscated were identified as dead, injured or grossly sick over a 10 day period, and some industry average mortality rates for amphibians have been estimated to be as high as 72% (Ashley et al., 2014). In addition, assessments of welfare conditions in exotic pet markets have found significant animal welfare problems throughout, with a high prevalence of stressed animals due to the routine handling, inappropriate thermal conditions, and inadequate temporary housing (Arena et al., 2012).

Pet owners or visitors to markets and pet shops may assume that the amphibians on sale are captive-bred and are adapted to live in captivity. However, there is often no legal requirement to state the origin of the species or whether they are wild-caught or captive-bred (Hughes, 2021). Furthermore, captive-bred amphibians are no more suited to captivity than their wild-caught counterparts (Grant et al., 2017; Warwick et al., 2017), because they still maintain their wild instincts and consequently may be treated and housed in conditions that fail to meet their needs (Can et al., 2019).

The keyword 'stress', which referred to the subjective state, returned the most results in our review overall (n = 119, 62.9% of all results). Identifying behavioural signs of stress in amphibians is challenging for the uninformed observer, and many amphibian pet owners may unknowingly cause mental stress, or allow their pet to be stressed for prolonged periods (Jacobson et al., 2021). Knowledge of the needs of amphibians is limited, and little is known about what constitutes good welfare (Brod et al., 2019; Ferrell, 2021). For example, it is currently not clear whether or not live feed is critical for the welfare of amphibians, particularly in terms of behavioural needs, or whether dried or frozen feed is adequate (Brod et al., 2019). Furthermore, amphibians are often kept in solitary housing, despite being social animals. Tungara frogs, for example, are known to exhibit complex social behaviours (Burghardt, 2013). Despite best intentions, many owners of pet amphibians keep them in enclosures that are too small, and at incorrect humidity and temperature levels, and many amphibians are left under-stimulated and given inadequate diets that can result in health issues (Burghardt, 2013; Grant et al., 2017; Warwick et al., 2018). Our findings highlight the fact that the barren, under-stimulating tanks in which pet amphibians are often kept in by the uninformed owner, are inadequate for these complex animals who are capable of a range of subjective states (Burghardt, 2013). Even the most well-intentioned amphibian pet owner is limited in their ability to assure the welfare of their pets, due to the lack of knowledge of amphibian needs in captivity, and so many amphibians suffer as a result (Grant et al., 2017).

4.4.1. Beyond the exotic pet trade

Amphibians are widely used and traded for consumption, traditional medicines, and scientific research (Gerson, 2012; Little et al., 2021). For example, the international trade in frogs' legs as food has grown substantially over the past 30 years, with Europe being the biggest importer of Indonesian frogs' legs for human consumption (Grano, 2020). Frogs traded as food are either wild-caught or captive-bred, and are subject to similar welfare issues as those destined for the exotic pet trade. Once at the market though, frogs are generally killed on the premises (Grano, 2020) by having their legs cut off with a knife or scissors, or are dismembered by hand (Altherr et al., 2011). Slaughtering frogs in this way is likely to cause extensive suffering and a prolonged death for the individuals involved. Amphibians used for scientific research may also experience considerable welfare issues. In addition to the issues associated with trade, such as wild-capture, captive breeding, transportation, and captivity (Baker et al., 2013; Hughes et al., 2021), amphibians used in research may be subjected to painful procedures, given noxious chemicals, or killed as 'waste' animals (Bailey, 2019; Prescott and Lidster, 2017). The number of amphibians used in scientific research has increased in recent years, yet knowledge of their welfare needs is still limited (Brod et al., 2019). Legislation protecting amphibians in scientific research varies around the world (Fry, 2012), and as such, so does the degree of suffering they may be subjected to (Beauchamp and Morton, 2015).

4.5. Limitations

Our review was intended to provide a snapshot of what is currently known about amphibian sentience in a portion of the scientific literature. By focussing on several journal databases, we hope to have covered the majority of the literature, but, due to the search method employed, we recognise that some potentially relevant studies, such as review articles, will have been omitted. Furthermore, our review is limited in terms of the amphibian search terms used, as time constraints meant that we could not search for every amphibian species individually. However, we hope that, by using the different taxonomic orders and the common names for amphibians, we have captured the vast majority of relevant studies. Caution must be taken when relating findings from experimental studies to the diverse and complex nature of international trade. But, as we know amphibians are sentient, more care and consideration of their welfare is needed.

4.6. Future research

Our review of amphibian sentience has only highlighted a limited number of traits and capacities that are relevant to amphibian welfare. Future research is sorely needed to expand our understanding of amphibian sentience and to ensure that this knowledge is used to improve amphibian welfare across all of the contexts and industries that affect and use them. Of the 10 keywords returned in our review, only one of them, 'emotion', was explored. Whereas, the capacity of amphibians to feel pain, stress and fear, among other states, have only been assumed and have not been subject to scrutiny in the portion of the scientific literature that we reviewed in the past 31 years. The apparent acceptance of these traits in amphibians is perhaps largely because they are a vertebrate species, and collectively there is bountiful evidence that vertebrates are capable of pain and suffering (see Proctor et al., 2013). However, there are still many keywords unmentioned in the literature, most notably positive states such as pleasure and play. This may be because there is still comparatively little research focussing specifically on amphibian welfare, behaviour and sentience, compared with other vertebrate taxa such as mammals (Proctor et al., 2013). Nevertheless, welfare science now recognises the importance of positive emotions and states in captive animals for ensuring good or even adequate welfare (Mellor, 2012). Therefore, it is fundamental that we learn more about both positive and negative emotions and states in amphibians, so that the needs of amphibians can be better catered for, and informed ethical judgements can be made regarding if and when amphibians should be exploited, or whether some practices or species should be restricted or prohibited entirely.

Whether amphibians are traded for exotic pets, consumption, or research, there are considerable welfare issues associated with the capture, restraint, captive breeding, transportation, and housing of these animals (Baker et al., 2013; Brod et al., 2019; Ferrell, 2021). Given that amphibians are sentient beings, and are capable of a range of emotional states, more needs to be done to ensure their welfare. Too little is known about whether or not amphibians can experience good welfare when traded and kept in captivity, or what is required to ensure their wellbeing. Currently, limitations or concerns regarding the species that can and cannot be traded tend to be focussed on their conservation status and include solutions ranging from complete bans to the creation of positive lists (Herrel and Van Der Meijden, 2014; Warwick et al., 2018). We recommend that the suitability of amphibians to captivity, and their needs in terms of welfare, should also be a major factor in determining whether species should be traded. Where trade continues, much more research is needed to ensure that we understand more about the mental lives of amphibians, and how to maximise their welfare throughout the trade chain, and ultimately in captivity.

5. Conclusion

Amphibians, along with other vertebrate taxa, are sentient beings, and their wellbeing matters. Our review has clearly shown that amphibians are considered capable of a range of feelings and states. The capacity of amphibians to experience negative states such as fear, pain, anxiety, and distress have important implications for their treatment in the wildlife trade, as well as in other contexts such as biomedical research. However, measuring emotions in amphibians is challenging, particularly as they do not use facial cues like mammals do, and we are often unable to accurately interpret their vocal and body posture cues (Burghardt, 2013). Our current lack of understanding of amphibian sentience does not indicate a lack of sentience in amphibians, it just means that we scientists have to be more creative in developing ways to explore the subjective minds of amphibians. The same issues are found in other taxa, including reptiles (Lambert et al., 2019) and insects (Lambert et al., 2021), who, like amphibians, also score poorly compared with mammals and birds on belief in animal mind studies (Callahan et al., 2021; Higgs et al., 2020). Yet, reptiles and insects are also subject to extensive use by humans, and as a result their welfare also deserves attention (Lambert et al., 2021, 2019). The field of animal sentience is still in its infancy, and as science and technology progress, we will learn new ways to understand amphibians. These findings can then be used to safeguard the welfare of the millions of amphibians who are traded each year.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.applanim.2022.105559.

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